WEEK 1 · LESSON 6 OF 24 · HANDOUT

Week 1 Project: Predict Rents in Kigali

Study time: **about 90 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=RlQuVL6-qe8 — open the lesson online to watch inside the course.

🎓 **Go deeper:** Data School: Comparing machine learning models in scikit-learn — https://www.youtube.com/watch?v=0pP4EwWJgIU

🎓 **Go deeper:** freeCodeCamp: scikit-learn — full course — https://www.youtube.com/watch?v=pqNCD_5r0IU

Your goal in this lesson: Run a complete regression project from raw data to a short report: prepare the features, compare three models against a baseline, and explain the winner.

The brief

A property website in Kigali wants to show visitors a **fair rent estimate** for any home. You are the data scientist. Build the model, test it honestly and write a one-page report for the manager.

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(7)
n = 600
district = rng.choice(["Gasabo", "Kicukiro", "Nyarugenge"], n, p=[0.45, 0.35, 0.2])
bedrooms = rng.integers(1, 5, n)
size_m2 = (bedrooms * 28 + rng.normal(20, 12, n)).clip(25).round()
km_to_centre = rng.uniform(0.5, 15, n).round(1)
furnished = rng.random(n) < 0.3
base = {"Gasabo": 120_000, "Kicukiro": 90_000, "Nyarugenge": 100_000}
rent = (np.array([base[d] for d in district]) + 2_600 * size_m2 - 6_000 * km_to_centre
        + np.where(furnished, 60_000, 0) + rng.normal(0, 25_000, n)).round(-3)
homes = pd.DataFrame({"district": district, "bedrooms": bedrooms, "size_m2": size_m2,
                      "km_to_centre": km_to_centre, "furnished": furnished, "rent_rwf": rent})
print(homes.shape)
homes.head()
```

Step 1 — Explore (15 minutes)

- `homes.describe()` and `homes["district"].value_counts()`.
- Average rent per district (`groupby`).
- A scatter plot of `size_m2` against `rent_rwf`, and one of `km_to_centre` against rent.

Write two sentences on what you see.

Step 2 — Prepare

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X = pd.get_dummies(homes.drop(columns="rent_rwf"), columns=["district"], dtype=int)
X["furnished"] = X["furnished"].astype(int)
y = homes["rent_rwf"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

scaler = StandardScaler().fit(X_train)
X_train_s, X_test_s = scaler.transform(X_train), scaler.transform(X_test)
```

Step 3 — Compare models

```
from sklearn.dummy import DummyRegressor
from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_absolute_error, r2_score

models = {
    "Baseline (mean)": DummyRegressor(),
    "Linear regression": LinearRegression(),
    "k-NN (k=10)": KNeighborsRegressor(n_neighbors=10),
}
results = []
for name, m in models.items():
    m.fit(X_train_s, y_train)
    pred = m.predict(X_test_s)
    results.append({"model": name,
                    "MAE": round(mean_absolute_error(y_test, pred)),
                    "R2": round(r2_score(y_test, pred), 3)})
print(pd.DataFrame(results))
```

Linear regression should win clearly (MAE about 20,000 RWF), k-NN comes second and the baseline is far behind. That makes sense: the rents were built from a straight-line rule, which is exactly what linear regression can learn.

Step 4 — Explain the winner

- Print the coefficients of a linear model trained on the **unscaled** features (they are easier to read in RWF).
- Which three features matter most? Explain them in plain words.
- Plot predicted vs real rent for the test homes.

Step 5 — Use it

Estimate the rent for three example homes of your choice (one per district). Present the answers rounded to the nearest 5,000 RWF — false precision looks unprofessional.

Step 6 — Report (one page)

1. **Goal** — one sentence.
2. **Data** — 600 simulated homes; the features used.
3. **Results** — a small table of the three models with MAE and R^2.
4. **What drives rent** — the top three features, in RWF.

5. **Limits** — at least two (for example: simulated data; no information on building quality, water or security; rents change every year).

Marking guide (self-check)

Area	Excellent work
Preparation	All features numeric, split before scaling
Comparison	Baseline included, same test set for every model
Explanation	Coefficients explained in plain RWF terms
Honesty	Limits stated clearly
Communication	A manager understands it in five minutes

Remember this: A real project is a story: explore, prepare, compare against a baseline, explain the winner and state its limits.

Name: _____ Date: _____

Activity

Task 1

Finish Steps 1–6 in one notebook. Then challenge yourself: remove the `district` columns and retrain. How much worse does the model get? What does that tell the property website?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Why compare every model with a baseline?
   A. To prove the model is better than a lazy guess
   B. To slow down training
   C. To make the table longer
   D. Because scikit-learn requires it
2. Why must all models be tested on the same test set?
   A. To save memory
   B. So the comparison is fair
   C. It is not important
   D. It is the only way to train
3. Why did linear regression win on this data?
   A. It is always the best model
   B. The rents were built from a straight-line rule
   C. It ignores the data
   D. It uses distances
4. Why train a separate linear model on unscaled features for the report?
   A. It is more accurate
   B. Its coefficients are in real units (RWF), easier to explain
   C. To leak data
   D. Scaled models cannot predict
5. Which is an honest limitation of this project?
   A. The data is simulated and lacks building quality and security
   B. R^2 is always 1
   C. The model is perfect
   D. There are no limits
6. Why round rent estimates to the nearest 5,000 RWF?
   A. To hide mistakes
   B. Avoid false precision
   C. Python requires it
   D. The model cannot add

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